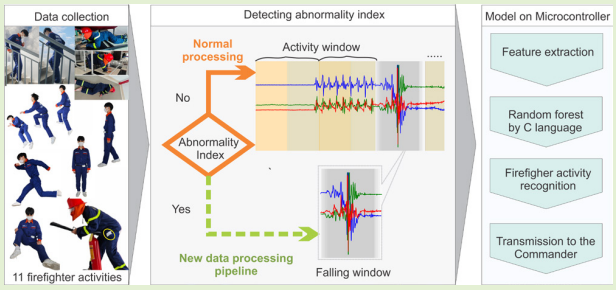
RFAR: A Real-Time Firefighter Activity Recognition System Using Wearable Accelerometer

To-Hieu Dao, Duc-Nghia Tran, Viet-Hoan Bui, Van Son Nguyen, Dang Khanh Hoa, Pham Van Thanh, and Duc-Tan Tran, *Member, IEEE*

***Abstract*—Human activity recognition (HAR) presents significant challenges. Although HAR is mainly used for elderly care, it holds significant potential for supporting firefighter operations. The main challenge of HAR is the recognition of complex actions and continuous transitions from real-time data. This study proposes a real-time firefighter activity recognition (RFAR) system that accurately recognizes 11 activities. The proposed system is based on a wearable device with an integrated accelerometer, ensuring compactness and energy efficiency. It can optimize parameters for deploying the random forest (RF) model on performance-constrained microcontrollers. A new data processing pipeline that integrates a sliding window technique with an abnormality index (AIx) detection technique is proposed. This method achieves an *F*1-score of 96.1% on a private dataset. Real-time experiments with a waist-mounted accelerometer achieved an *F*1-score of 96.2% in rescue simulations conducted by volunteers at a firefighting training facility in Vietnam. To validate, the method is performed on three public datasets: achieving an *F*1-score of 96.6% on SFDLA, 96.6% on MobiFall, and 78.2% on UniMiB-SHAR. Another contribution of this study is the development of a specialized dataset that comprehensively represents fundamental and unique firefighter activities in realistic scenarios.**



***Index Terms*— Accelerometer, activity, firefighter, real-time, real-time firefighter activity recognition (RFAR), recognition, wearable device.**

I. Introduction

RAPID urbanization and gradual deterioration of infrastructure have contributed to increased latent risks of fire and explosion in both residential and industrial zones [1]. Uncontrolled construction activities and poor management have left many buildings vulnerable to safety hazards [2]. Meanwhile, many local fire protection systems have become outdated or insufficient to meet current operational demands [3]. These factors have led to a rise in the number of fire incidents, which tend to exhibit greater severity [2].

Firefighters play a central role in emergency response operations and frequently face potential hazards [4]. Common hazards include extreme heat [5], toxic gases [6], confined spaces [7], and the potential for structural collapse [8]. In the United States, a total of 23 075 rescue workers were injured according to a report on firefighter fatalities during 2018–2020[1] period. The majority of these injuries, accounting for 87%, occurred during structural fires, where the likelihood of permanent injuries is ten times higher than in nonstructural fires. The NFPA statistics[2] estimated 63 175 injuries, with 89 on-duty fatalities reported in 2023. Fire suppression activities posed the highest risk, accounting for approximately 18 875 cases. A significant challenge faced by firefighters is the lack of tools and solutions to support them in their duties [9], [10], [11], [12].

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[1] https://www.usfa.fema.gov/statistics/ (Accessed: 12:00 25/01/2025).

[2] https://www.nfpa.org/ (Accessed: 8:00 25/01/2025).

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The application of science and technology significantly enhances firefighter safety [4], [7], [9]. Recent research has focused on three main areas of improvement: upgrading the protective gear to mitigate the effects of heat and toxic chemicals [6], [10]; monitoring psychological and health status under high-stress working conditions [3], [5]; and developing on-site monitoring and communication systems to assist decision-making [4], [11], [12]. Among these, communication systems between command units and frontline firefighters are critical in maintaining continuous communication for operational coordination [13], [14].

Monitoring of firefighters' performance and their communication with commanders are vital for rescue personnel's timely response and risk mitigation to [4] and [7]. However, traditional communication methods such as two-way radios often provide fragmented information and lack the ability to monitor the real-time status of individual personnel [15]. This significantly hampers rescue coordination, especially when the location or condition of a firefighter cannot be determined—such as in cases of unconsciousness or immobility [16]. Delayed responses in such situations may lead to severe consequences [9], [17], [18].

The lack of accurate and continuous information on each firefighter's position and conditions poses major challenges for command and rescue coordination [5], [12], [19]. Human activity recognition (HAR) for firefighters, combined with location information, is a potential solution that provides objective and continuous data, supporting timely decision-making [20]. HAR is commonly approached through two primary methods: computer vision [21] and wearable sensor signals [22]. However, Minh Dang et al. [21] identified the main drawbacks of vision-based activity recognition methods, including sensitivity to environmental conditions, complex data processing requirements, and privacy concerns due to continuous recording. In contrast, wearable sensors are highly appreciated for their outstanding advantages, such as compact size, low cost, and the ability to quickly detect kinematic changes [11], [22], [23], [24], [25], [26], [27], [28], [29], [30].

Table I provides an overview of recent studies in HAR, using wearable devices equipped with accelerometers. They are typically placed at positions such as the waist [26], [31], wrist [32], or thigh [11] to collect activity data. Many studies have integrated accelerometers with other sensors, such as gyroscopes [24], carbon monoxide sensors [33], or nine-axis inertial sensors [29] to enhance the detection of complex movements. In addition, electromyography sensors [30] have been used to capture detailed muscle signals.

Accelerometer-based HAR is commonly used in health monitoring [31], [34], [35], [36], [37], [38], [39], [40], but it also holds potential for real-time activity tracking and incident detection in firefighters [41], [42], [43]. Most studies primarily focus on distinguishing fall situations [3], [11], [29], [30], with little attention given to specific firefighter activities [29], [30] such as crawling and slithering. These activities are performed in harsh environments with high temperatures, thick smoke, and confined spaces [44], making recognition challenging. The main challenges stem from the similarity of postures, the difficulty in controlling sensor data, and their continuous real-time transitions. Nevertheless, recent studies have applied deep learning (DL) [30], [37], [45], [46] to improve recognition accuracy. These approaches typically require high computational resources, making it challenging to implement them on microcontrollers with limited memory, processing speed, and energy capacity. High energy consumption shortens operational time, necessitates larger batteries, and limits the design of compact wearable devices for firefighters. To address this, adopting lightweight machine learning (ML) models deployable on microcontrollers offers a practical solution for optimizing resources while maintaining a balance between accuracy, processing speed, and power efficiency.

TABLE I
OVERVIEW OF HAR USING WEARABLE DEVICES

Year	Dataset	Research object	Methods
2024	Accelerometer ; UniMiB (50 Hz), WISDM (20 Hz), UCI (50 Hz); body: front pocket.	Human, activity	CLS-CNN-ResNest [47]
2024	Fusion sensors; 100 Hz; 18 volunteers; body: arm, upper arm, calf; 8 activities	Firefighter, activity	Hybrid ML [30]
2023	Accelerometer; Opportuni (30 Hz), UniMiB (50 Hz), UR Fall (256 Hz), UCI (50 Hz); body: front pocket.	Human, activity	CNN-DCT, ResNet-DCT [37]
2022	Accelerometer, CO; 100 Hz; 8 volunteers; body: waist; 2 activities	Firefighter, falling	Fall detection [33]
2021	9-axis[1], 15 Hz; 10 volunteers; body: chest, elbow, waist, thigh, ankle; 2 activities	Firefighter, falling	LSTM [29]
2021	Accelerometer; WISDM (20 Hz), PAMAP2 (33.33 Hz), UniMiB (50 Hz), UCI (50 Hz); body: front pocket.	Human, activity	HARMamba [48]
2020	Accelerometer; 50 Hz; 21 volunteers body: wrist; 6 activities	Human, activity	GBDT, RF, KNN, SVM [32]
2019	9-axis, CO; barometer; 100 Hz; 6 volunteers; body: front trouser, pocket; 2 activities	Firefighter, falling	Fall detection [11]
2019	Accelerometer; 50 Hz; 16 volunteers; body: waist; 2 activities	Elder, falling	Fall detection [31]
2016	On-body radio frequency, doppler; 20 Hz; body: chest, forehead, right wrist, right ankle; 7 activities	Firefighter, activity	SVM [49]

This study makes the following contributions as listed below.

1) First, a dedicated dataset was constructed to simulate firefighter activities during rescue training drills, aiming to supplement the gaps in the existing public datasets.
2) Second, this study developed a new signal processing pipeline that combines a sliding window technique with abnormality index (AIx) detection to enhance the quality of activity information at each stage. This pipeline thoroughly considers the transition process between actions and is suitable for deployment on resource-constrained microcontrollers. This aspect has received little attention in previous studies, which primarily focused on homogeneous and continuous actions.
3) Third, this study introduces the real-time firefighter activity recognition (RFAR) system, which uses a wearable device with a waist-mounted accelerometer

[3]9-axis: accelerometer, gyroscope, magnetic.

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to recognize 11 firefighter-specific activities, including walking (WA), standing (ST), lying (LY), sitting (SI), jogging (JO), walkstoop (WS), crawling (CR), slithering (SL), falling (FA), downstairs (DS), and upstairs (US). The system is implemented on a low-power microcontroller using a random forest (RF) model and a feature set, both of which have been optimized. It achieves high accuracy while keeping the model size under 1 MB, making it suitable for deployment on resource-constrained embedded platforms.

4) Fourth, the developed wearable device is lightweight (under 200 g), enables continuous operation for up to 24 h, and is designed for portability and comfort of users on the scene. The system was evaluated on volunteers from Phenikaa University and the University of Fire Prevention and Fighting in Vietnam, showing consistent results with over 96% $F1$-score and 99% accuracy in realistic simulated rescue scenarios. The system was also tested on public datasets including SFDLA, MobiFall, and UniMiB-SHAR, where it yielded over 78% $F1$-score and over 97% accuracy.

II. Related Works

A. Activity Recognition for Elderly Individuals

The aging population has led to an increased demand for elderly healthcare [26], [31], [36], [38], [39]. Research on HAR provides essential input for alert systems, supporting the early detection of health issues [24], [26]. To meet the requirements of continuous and minimally invasive monitoring, wearable devices are widely used due to their ability to collect real-time motion data at low cost [22], [25], [50]. The advancement of artificial intelligence (AI) has further enhanced the automatic analysis of HAR data, with ML [19], [51] and DL [37], [46], [47], [52], [53], [54] being the two primary approaches.

Traditional ML is widely applied in HAR due to their simplicity, suitability for deployment on low-power wearable devices, and their ability to balance cost, energy efficiency, and recognition performance [22], [24]. Thu et al. [26] used a decision tree (DT) to classify five basic activities (LY, SI, ST, WA, and JO), achieving 99% accuracy on an ESP8266 microcontroller. Hong et al. [22] implemented a system integrating components such as an accelerometer, a low-power microcontroller, and RF, achieving an accuracy of 99.2% in recognizing four basic states: ST, SI, JO, and WA. Lu et al. [32] proposed an HAR method using a single accelerometer by grouping activities and combining global and local features, achieving 93% accuracy with gradient boosting DT (GBDT), RF, k-nearest neighbor (KNN), and support vector machine (SVM) algorithms. Unlike Lu's method, Wang et al. [35] applied a 2.56-s window with 50% overlap and used a hierarchical tree-based classification structure. RF and SVM models were applied at each node with optimized feature selection, yielding 90.4% accuracy using both accelerometer and gyroscope data.

In addition to time-domain features, frequency-domain features were also explored to enhance classification performance. Preece et al. [55] compared 14 feature types from accelerometer signals across time, frequency, and wavelet domains. Using a 2-s time window with 50% overlap and applying KNN, the classification of eight activities achieved a maximum accuracy of 94% when using fast Fourier transform features from an ankle-mounted sensor.

DL has demonstrated superior performance in HAR due to their ability to automatically learn features from sensor signals and model complex activity patterns [45]. In a study [54], a triple cross-domain attention mechanism across sensor, temporal, and channel dimensions was introduced to improve HAR performance on convolutional neural network (CNN) and residual network (ResNet). It improved accuracy by 0.96%–1.47% on the UniMiB-SHAR dataset without increasing the number of parameters. Xu et al. [37] proposed a channel attention mechanism based on discrete cosine transform (DCT) in the frequency domain to improve model performance in HAR tasks. By leveraging representative frequency components, the method enabled the ResNet model to better learn informative features of each activity and achieved an accuracy of 79.51%, compared to 76.38% with the baseline model. Qin et al. [47] proposed a collaborative learning strategy CNN and ResNest model (CLS-CNN-ResNest) that enables simultaneous adjustment of network width and input resolution, allowing a balance between accuracy and computational cost. The model achieved 99.12% accuracy on WISDM, 97.64% on UCI-HAR, and 78.91% on UniMiB-SHAR and supports on-device inference without retraining through a weight-sharing mechanism among subnetworks. In a different approach, Li et al. [46] focused on smart home environments with multiple users and distributed environmental sensors. The proposed HAR on wide time-domain CNN (WCNN) model integrates sensor data contribution significance analysis and a spatial distance matrix to enhance recognition accuracy and reduce noise caused by overlapping activities. Experiments on the CASAS dataset showed that the model achieved an accuracy of 86.68%–97.08%, while significantly reducing processing time compared with long short-term memory (LSTM).

DL offers high accuracy through automatic feature learning but typically requires substantial computational resources, making them less suitable for resource-constrained embedded devices. For real-time HAR applications, especially on low-power microcontroller platforms, traditional ML remains a more practical option due to its simplicity, low cost, and efficient deployment.

B. Firefighter Activity Recognition

FAR systems address dynamic, high-intensity movements in hazardous environments. These systems aim to enhance safety and efficiency by monitoring physical activities, detecting anomalies, and providing real-time information during emergency situations [5], [11], [42], [43].

Geng et al. [49] introduced a firefighter activity recognition method based on radio frequency (RFY) signals captured via wearable sensors. They used an SVM with a Gaussian kernel to recognize seven activities: ST, WA, JO, LY, CR, US, and DS. The system achieved an average accuracy of 88.7%, with static activities such as ST and LY being classified more accurately, while dynamic activities such as JO, US, and DS were more prone to misclassification due to similar RFY

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features. Wrist and ankle sensor placements yielded relatively better results for dynamic movements. However, the use of a 10-s time window limited the system's responsiveness to fast, complex firefighter actions and introduced additional noise.

A support system incorporating sensor fusion algorithms was developed by Pham et al. [11] to detect falling, monitor physical incapacitation, and assess environmental conditions. The falling detection algorithm combined data on acceleration, posture, and altitude, achieving 100% accuracy on experimental datasets and 97.9% accuracy on public datasets. The system also monitored CO levels and issued alerts when concentrations exceeded 33 ppm, while barometric data helped differentiate between physical falls and elevator use.

Following a similar approach, Dang et al. [33] developed a real-time support system for firefighters, using an accelerometer and a CO sensor to detect falling and monitor CO concentration. The accelerometer sampled at a frequency of 100 Hz to recognize body states (FA, ST, WA), while the CO sensor measured gas concentration at 10 Hz to detect levels exceeding 35 ppm. The system sent alerts to supervisors via a GSM/GPRS module. The results showed a fall detection accuracy of 93%, a sensitivity of 96.5%, and an overall performance of 99.4% on public datasets.

Using a different approach, Chai et al. [29] developed a smart fall detection system for firefighters, integrating sensors into personal protective clothing at five body locations. The system used sensor fusion techniques and a recurrent neural network with three layers of LSTM to recognize fall events and fall-like activities. The system indicated an accuracy of 94.1%, a sensitivity of 92.25%, and a specificity of 94.59%, with the chest sensor playing a critical role in improving detection performance.

Taking a sensor-integrated clothing approach, Chai et al. [30] examined the physical training routines of firefighters by assessing oxygen consumption following at least 20 test sessions. Their findings revealed that some participants showed signs of fatigue or suboptimal physical condition, as reflected in their tendency to choose ST over SI during rest intervals. Despite offering insights into physical readiness, such training protocols are designed for controlled environments and may not fully capture the complexity and unpredictability of real emergency scenarios.

Most studies using wearable sensors primarily focus on fall detection for firefighters. Furthermore, there are very few publicly available datasets for studying and analyzing their activities. Therefore, in this study, a comprehensive dataset of firefighter behaviors, including falls, has been developed to evaluate the system. This study aims to develop a real-time system for recognizing 11 fundamental firefighter activities during search and rescue missions. Current studies still have many limitations, such as recognizing only a limited number of activities and lacking real-time processing capabilities. The proposed system uses an RF algorithm embedded in a low-power microcontroller to process acceleration data collected from a waist-mounted sensor. The goal is to provide a highly accurate solution while optimizing size, weight, and power consumption. In addition, the system will integrate firefighter location information within the building to assist commanders in monitoring real-time status. This enables timely interventions in hazardous conditions such as falls or unconsciousness. The study focuses on developing the RFAR system to recognize specific activities commonly performed during firefighter missions, including SL, CR, WS, and FA.

TABLE II
HARDWARE CONFIGURATION AND EXPERIMENTAL PLATFORM

Module	Component	Specification
Wearable device	ADXL345	3-axis accelerometer
	[1]ESP32	520 KB RAM, 448 KB ROM, 4 MB flash
	Micro SD card	2 GB
	Communication	433 MHz; 8000 meters; 2.4 Kbps
	Battery	Li-ion, 2000 mAh
Central device	Communication	433 MHz; 8000 meters; 2.4 Kbps
	ESP32	520KB RAM, 448KB ROM, 4 MB flash
	Power	Adapter 12 VDC —2A
Server	Chipset	Intel Xeon Silver 4310; 2.1 GHz speed
	RAM	16 GB RDIMM
	Hard disk	2TB IDD NLSAS
	Software	Visual studio code
	Platform	Windows server 2019
	Protocol	Message queuing telemetry transport

[1] https://www.espressif.com/

III. RESEARCH METHOD

This study developed an activity recognition system optimized for deployment on a low-power microcontroller to process real-time data. The process involved detailed system modeling, data collection, feature extraction, and model training. The modeling phase defined the system's structure and functionality, while practical data collection ensured reliability and representativeness. The main features were extracted to optimize recognition performance before training the model to achieve high accuracy.

A. System Architecture

Fig. 1 illustrates the overall architecture and operation flow of the proposed FAR system. This system is designed to monitor and assist commanders in tracking the operational status of each team member. Each firefighter is equipped with a compact wearable device mounted at the waist, capable of recognizing and transmitting real-time activity data to a central device controlled by the commander. This central device visually displays the activity status of team members, aiding in timely decision-making during emergency situations.

The system is capable of recognizing a wide range of activities including WA, ST, LY, SI, JO, WS, CR, SL, FA, DS, and US. The live data corresponding to each activity are plotted in real-time and illustrated in Fig. 1. It uses an RFY communication module to ensure low-latency data transmission, enabling near-instantaneous updates to the central station. Powered by a 2000-mAh LiPo battery, the device supports extended operations while minimizing the need for frequent recharging. In addition, a low-power microcontroller processes sensor data and manages communication functions, maintaining high performance with optimized energy efficiency. The system configuration used in this study is described in Table II.

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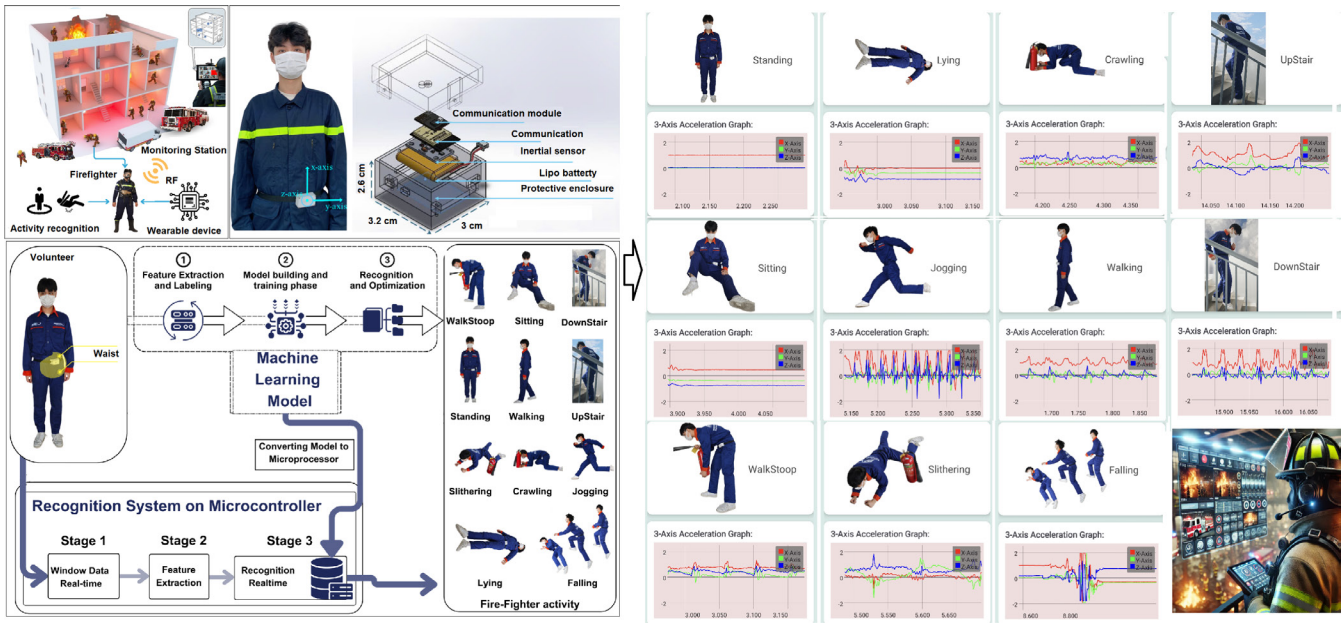

Fig. 1. System for recognizing activities of firefighters through a waist-worn device, interacting with the command screen via RFY wave.

The system operates in three main stages: real-time data collection through a sliding window, feature extraction, and activity recognition. The extracted features are used to train the ML, which is subsequently optimized and deployed on a microcontroller integrated with the device. This system enables the rapid detection and reporting of critical activities, supporting commanders in monitoring and coordinating team operations in hazardous environments. By enhancing both safety and efficiency, the system is particularly suited for degraded buildings where prior system installation is not feasible and can operate reliably in fire conditions, even during power outages.

B. Dataset

This study uses four datasets on human activities, including one private dataset focused on firefighter-specific actions and three public datasets that contain similar human actions.

1) Private Dataset: The firefighter activity dataset was collected from a group of 20 male volunteers, all in good health, between 22 and 35 years of age, with heights ranging from 1.6 to 1.8 m and weights from 60 to 80 kg. These volunteers were carefully selected from Phenikaa University and the University of Fire Prevention and Fighting, with no female participants included to ensure uniformity in physical characteristics. Data were collected at a sampling frequency (F_s) of 100 Hz and comprised three-axis accelerometer measurements recorded in units of g ($g = 9.8$ m/s^2), captured during practical rescue and firefighting drills. Complex activities were sampled more frequently to ensure objectivity in dataset construction. The detailed description and the number of samples for each activity in the private dataset are presented in Table III. Each activity was performed continuously for at least 3 min, except for the FA activity, which was conducted within 3–5 s (seconds). The collected data were stored on a microSD card and recorded at the waist–hip area of the volunteers to ensure accuracy and signal stability. The total data collection time amounted to 371 min, with the dataset size exceeding 100 MB. In this study, parameter optimization for the RF model was conducted on a private dataset, while DT, GBDT, SVM, light gradient boosting machine (LBM), and extreme gradient boosting (XGB) models were used to objectively evaluate the effectiveness of the optimization process.

TABLE III
DESCRIPTION OF FIREFIGHTER ACTIVITIES ON PRIVATE DATASET

Activity	Description	Samples
WA	Moving upright maintaining at least one foot on the ground.	123,932
ST	Maintaining an upright posture without movement.	56,016
LY	Positioning the body on a flat surface with the back touching the ground	82,516
SI	A resting posture with the buttocks and feet in contact with the ground.	63,013
JO	A fast movement state with running steps and no more than one foot on the ground at a time.	90,783
WS	Slow walking in a bent or crouched posture.	83,242
CR	Moving forward on hands and knees, keeping the torso slightly elevated off the ground.	82,669
SL	Moving close to the ground on hands and knees, with the body parallel to the ground.	90,340
FA	Sudden loss of balance resulting in contact with the ground.	129,815
DS	Descend the stairs on foot at a moderate pace.	151,448
US	Climb the stairs on foot at a moderate pace.	157,882

2) Public Dataset: In this study, a new approach is applied, in which falling is considered a behavior rather than merely an isolated event. This perspective differs from previous studies, which often focused separately on either activity recognition or fall detection [12], [31], [39]. Therefore, three publicly available datasets, [3]MobiFall [56], [4]SFDLA [40], and [5]UniMiB-SHAR [57] containing similar activities, are used for comparison to ensure the diversity and objectivity of the study.

[3]https://www.kaggle.com/datasets/kmknation/mobifall-dataset-v20
[4]https://kilyos.ee.bilkent.edu.tr/ billur/
[5]http://www.sal.disco.unimib.it/technologies/unimib-shar/

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C. Data Processing

1) Segmenting Data for Nonfall Activities: Assume that an activity signal sequence $\overline{s}[n]$, composed of accelerometer data $\overline{s}[n] = (\overline{a}_x[n], \overline{a}_y[n], \overline{a}_z[n])$, is given, where $\overline{N}$ is the length of the signal sequence $\overline{s}[n]$ and $n \in [1, \overline{N}]$. Here, $\overline{a}_x[n]$, $\overline{a}_y[n]$, and $\overline{a}_z[n]$ represent time-series signals on the respective x-, y-, and z-axes, obtained from the accelerometer sensor. $\overline{s}[n]$ was recorded by an accelerometer mounted on the firefighter's waist during their activities.

Sampling frequency plays an important role in HAR system design. For wearable devices targeting elderly users, a low frequency of 1 Hz helps save energy and matches the slower nature of their movements [26]. In contrast, fast activities require higher frequencies between 50 and 100 Hz to capture more detailed data [22], [25], [42], [58]. Selecting an appropriate sampling rate helps balance recognition accuracy, response speed, and hardware constraints. However, processing data at high frequencies requires significant energy and computational resources. In this study, the sampling rate was reduced to 50 Hz to optimize processing efficiency while maintaining the necessary signal quality. To achieve this, a moving average filter was applied, where the average of every two consecutive samples was computed. For each axis $k \in \{x, y, z\}$, the filtered and downsampled signal $s[n]$ is calculated by averaging two consecutive samples from the original signal

$$a_k[n] = \frac{\overline{a}_k[2n-1] + \overline{a}_k[2n]}{2}, \quad n \in [1, N]. \tag{1}$$

Here, k denotes the x-, y-, and z-axes, while $s[n]$ is the downsampled signal with size $N = \overline{N}/2$, where $\overline{N}$ is the original signal size. F_s of $s[n]$ is 50 Hz, as applied throughout this study.

To analyze and classify various activities, the continuous signal $s[n]$ was segmented into smaller, equally sized time windows. Window sizes such as 10-s [22], 8-s [25], and 6-s [26] may cause delays in real-time responses and mix multiple actions within a single window. Previous studies on fall detection for firefighters [11], [33] and the elderly [37], [52], [59] have also used a 3-s window to achieve a balance between responsiveness and recognition accuracy.

This segmentation was performed by selecting a fixed time duration ($\Delta t = 3$ s) for each window and dividing the total duration of the recorded signal into time windows ($W_j[n]$). Each $W_j[n]$ contains M samples of $a_x[n]$, $a_y[n]$, and $a_z[n]$, corresponding to the time interval $[n_i, n_{i+1}]$, where n_i is the start time of the jth window and n_{i+1} is the end time, marking the point where the next window begins. The duration of each time window is given by $\Delta t = n_{i+1} - n_i$. Let the signal be segmented into $N_{\delta t}$ windows. Within each window $W_j[n]$, the accelerometer is represented as: $W_j[n] = (a_x[n], a_y[n], a_z[n]), \quad n \in [n_i, n_{i+1}], \quad j \in [1, N_{\delta t}]$. All the segmented activity data were divided into training and testing sets with a 60/40 split.

D. Labeling

The data labeling process involved segmenting the accelerometer data into windows and assigning labels corresponding to specific firefighter activities. Each activity was performed naturally by the volunteers in simulated scenarios designed to closely reflect practical conditions. The actions were directly observed and manually labeled by two supervisors to ensure annotation accuracy. The steps are as follows.

Step 1: There are 11 activities (WA, ST, LY, SI, JO, WA, CR, SL, FA, DS, and US), each corresponding to a unique label from 0 to 10.

Step 2: Sequential segmentation and labeling.

Dataset is processed sequentially for each action. For activity k ($k = 0, 1, \ldots, 10$), the data are segmented into $N_{\delta t}$ time windows and labeled before moving on to the next activity.

Step 3: Label assignment. For each window $W_j[n]$ belonging to activity k, the label L_j is assigned as $L_j = k$. Labels are assigned sequentially. After labeling all the windows for activity k, the process continues with activity $k+1$, where the new label is assigned as $L_j = k + 1$. The labeled dataset consists of pairs of windows and their corresponding labels, denoted as $\{(W_1[n], L_1), (W_2[n], L_2), \ldots, (W_{N_{\delta t}}[n], L_{N_{\delta t}})\}$. Here, L_j is determined by the activity associated with window $W_j[n]$. Labels are only updated after completing the labeling process for all data of the current activity. This approach ensures that each window $W_j[n]$ is linked to a single activity and assigned a unique label L_j.

1) New Data Processing Pipeline for Falling Detection: The above time windows were used only to recognize routine, repetitive activities, while the fall activity was segmented separately, based on detecting abnormal points that exceeded a threshold to create a signal region containing this activity. Fig. 2(a) illustrates a newly proposed pipeline for abnormality detection in data processing. The notable differences are highlighted in dark blue, representing the new process. In contrast, the green arrows depict the conventional process used previously. This approach enhances the detection of motion and fall actions while improving accuracy. This approach is designed to detect motion and fall actions while improving accuracy. A key highlight of the detection process is the modification of how data windows are generated when a fall event occurs. Specifically, instead of using a fixed sliding window, the system creates a specialized data window that includes the abnormal region ($s[n_0 - 3 \cdot F_s, n_0 + 3 \cdot F_s]$) around the detected fall time n_0. This approach focuses on the critical region and improves the performance of motion recognition and enhances the effectiveness of fall detection compared to conventional data processing methods. The abnormality detection method is carried out through the following steps.

Step 1 (Detecting the AIx): The AIx represents sudden variations in the accelerometer signal $s[n]$ at a specific moment. In this study, AIx is used to identify the moment when the firefighter falls and makes contact with the ground. To calculate AIx, the accelerometer signal $s[n]$ is transformed into a variation sequence $\Delta s[n]$, which quantifies the combined deviations of the three acceleration components $(a_x[n], a_y[n], a_z[n])$ from their respective means over a defined time mini window ΔW. Thanh et al. [31] indicated that the minimum time for the flight of fall period, from the onset of the fall until the body initially reaches the ground, is approximately 0.34 s. Thus, this study adopts $\Delta W = 0.5$ s as the minimum time for detecting AIx. The signal transformation process and the

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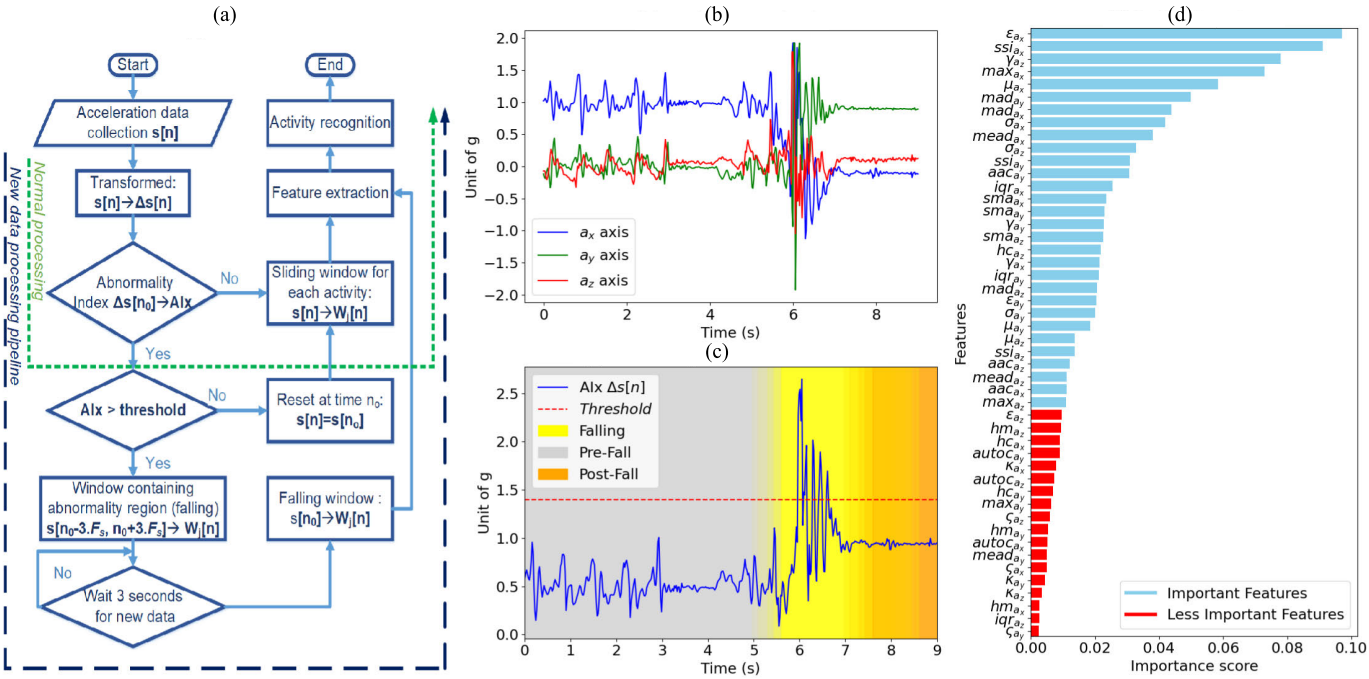

Fig. 2. (a) Flowchart of the abnormality detection and data processing pipeline, (b) signal variation during FA activity, (c) abnormality detection results, and (d) feature importance for activity recognition.

detection of areas corresponding to fall-related phases based on the AIx are illustrated in Fig. 2(b) and (c)

$$\Delta s\,[n] = \sqrt{\sum_{k\in\{x,y,z\text{axes}\}} \left(a_k\,[n] - \frac{1}{N}\sum\nolimits_{n=1}^{N} a_k\,[n]\right)^2}. \quad (2)$$

AIx is detected if $\Delta s[n_0]$ at time n_0 exceeds a predefined threshold

$$\text{AIx} = \begin{cases} \Delta s\,[n_0], & \text{if } \Delta s\,[n_0] > \text{threshold} = 1.8\,g, \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

where threshold is the upper threshold value proposed by Pham et al. [11].

Step 2 (Collecting Data Around the Abnormal Event): After detecting an abnormality at time n_0 from the accelerometer signal, additional data are collected for 3 s before and after this event $W_j[n]$ with $n \in [n_0 - 3 \cdot F_s, n_0 + 3 \cdot F_s]$.

Step 3 (Processing the Collected Data): The data collected during this time window are used to analyze and determine the severity of the fall event. The analysis process may involve extracting features from the accelerometer signals, such as the mean, standard deviation, or characteristic frequency of the signal, for the purpose of detecting falls [Fig. 2(c)].

E. Feature Extraction

Feature extraction is essential in recognizing activity [22], [25], [60]. In the study by Chai et al. [30], FAR was used to evaluate the effectiveness of training and physical fitness exercises for firefighters, achieving an accuracy of over 90%. They investigated 654 features extracted from surface electromyography, accelerometer, and gyroscope data for firefighting activity recognition (25 time and six frequency-domain feature types). To optimize computation, only low-complexity time-domain features for accelerometers are considered in this study, following the feature selection approach proposed by Chai et al. [30]. However, autoregressive coefficients involve computationally expensive operations, making it unsuitable for microcontroller implementation. Therefore, the selection from 16 time-domain feature types is evaluated and optimized, including energy (ε), mean absolute deviation (mad), signal magnitude area (sma), mean (μ), standard deviation (σ), simple square integral (ssi), average amplitude change (aac), maximum (max), interquartile range (iqr), Hjorth mobility (hm), median absolute deviation (mead), range (γ), skewness (ς), kurtosis (κ), autocorrelation (autoc), and Hjorth complexity (hc).

To reduce the number of features and optimize the performance of the recognition model, features with low importance were removed to decrease model complexity, while maintaining accuracy. A study [60] proposed an improved method for RF by gradually eliminating less important features. This approach identifies and removes irrelevant features during model training, leading to improved classification accuracy. Similarly, research [61] suggested using the standard deviation of the information gain values generated by each feature as a threshold. Features with values equal to or above this threshold are retained, while those below are discarded. In Fig. 2(d), the standard deviation of importance scores is approximately 0.02, indicating the spread of feature importance values. Therefore, a threshold of 0.01, chosen as half the standard deviation, was applied to eliminate features with minimal importance, effectively reducing the number of features while preserving model performance. As a result, 30 optimal features were selected, and their details are summarized in Table IV.

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TABLE IV
SUMMARY OF INFORMATION DESCRIBING THE FEATURES USED

Feature types	Axis	Description	Formula
Energy	x, y	The square root of the average of the squared signal values on x, y axes	$\varepsilon_{a_k} = \sqrt{\frac{1}{M}\sum_{n=1}^{M} a_k^2[n]}$
Mean absolute deviation	x, y, z	Mean deviation from the average value on x, y, z axes.	$\text{mad}_{a_k} = \frac{1}{M}\sum_{n=1}^{M} \lvert a_k[n] - \mu_{a_k}\rvert$
Signal magnitude area	x, y, z	Sum of absolute magnitude of signal on x, y, z axes.	$\text{sma}_{a_k} = \sum_{n=1}^{M} \lvert a_k[n]\rvert$
Mean	x, y, z	Average value of the signal on x, y, z axes.	$\mu_{a_k} = \frac{1}{M}\sum_{n=1}^{M} a_k[n]$
Standard deviation	x, y, z	Signal dispersion around the mean on x, y, z axes.	$\sigma_{a_k} = \sqrt{\frac{1}{M}\sum_{n=1}^{M} (a_k[n] - \mu_{a_k})^2}$
Simple square integral	x, y, z	Sum of the squared values of the signal on x, y, z axes.	$\text{ssi}_{a_k} = \sum_{n=1}^{M} a_k[n]^2$
Average amplitude change	x, y, z	Mean between consecutive samples on x, y, z axes.	$\text{aac}_{a_k} = \frac{1}{M-1}\sum_{n=1}^{M-1} \lvert a_k[n+1] - a_k[n]\rvert$
Maximum	x, z	Maximum value of the signal on x, z axes.	$\max_{a_k} = \max(a_k[n])$
Median absolute deviation	x, z	Median deviation from the median value on x, z axes.	$\text{mead}_{a_k} = \text{median}\left(\lvert a_k[n] - \text{median}(a_k)\rvert\right)$
Interquartile range	x, y	Difference between the 75th and 25th percentiles on x, y axes.	$\text{iqr}_{a_k} = Q_3(a_k[n]) - Q_1(a_k[n])$
Hjorth complexity	z	Measures the mobility of the signal on z axis: $\text{hc}_{a_k} = \frac{\text{hm}_{\Delta a_k}}{\text{hm}_{a_k}}$ *with*	$\text{hm}_{a_k} = \sqrt{\frac{\frac{1}{M-1}\sum_{n=1}^{M-1}(a_k[n+1] - a_k[n])^2}{\frac{1}{M}\sum_{n=1}^{M}(a_k[n] - \mu_{a_k})^2}}$
Range	x, y, z	Range between max and min values on x, y, z axes.	$\gamma_{a_k} = \max(a_k[n]) - \min(a_k[n])$

F. Optimization of RF Parameters on Microcontrollers

Suppose dataset $\mathcal{D} = \{(\text{feat}_j, L_j)\}_{j=1}^{N_{\delta t}}$, where

1) $L_j \in \{1, 2, \ldots, 11\}$ is the output label of feat_j, with 11 labels corresponding to different activities.
2) *RF Structure:* the model consists of n_estimatorsDTs, denoted as $T_1, T_2, \ldots, T_{n_\text{estimators}}$, with each tree limited to a maximum depth of max_depth. Each DT is built from a different subset of the training data $\mathcal{D}$ using the bootstrap sampling technique. Each tree T_e is trained on a subset $\mathcal{D}_e \subset \mathcal{D}$ and makes predictions for specific inputs.
3) *Prediction Using RF:* For an input feature vector $\text{feat}_j \in \mathcal{D}$, each tree in the forest makes a prediction

$$T_e\left(\text{feat}_j\right) \in \{1, 2, \ldots, 11\}, \quad e = 1, 2, \ldots, n_\text{estimators}$$

where $T_e(\text{feat}_j)$ is the predicted label of the eth tree.
4) *Final Prediction:* The final predicted label is determined by majority voting over the predictions of all trees

$$\hat{L}_j = \text{Voting}\left(T_1\left(\text{feat}_j\right), T_2\left(\text{feat}_j\right), \ldots, T_{n_\text{estimators}}\left(\text{feat}_j\right)\right).$$

RF [24], [25] with optimized weights is a suitable solution due to its compact size and low inference latency. However, RAM on microcontrollers is typically constrained, as it must concurrently support real-time data processing and system-level operations, while flash memory is occupied by firmware, sensor buffers, and configuration data [50]. As a result, the available capacity for deploying tinyML is limited [62], and recent studies [63], [64], [65] have typically deployed models on microcontrollers with approximately 1 MB of nonvolatile memory.

Algorithm 1 describes the hyperparameter tuning process for the RF model, based on an $F1$-score micro ($F1_{\text{mi}}$) and the constraint that the model size must not exceed 1 MB. To meet the requirement of optimizing the model for deployment on memory-constrained microcontrollers, two key hyperparameters, n_estimators and max_depth, were carefully adjusted. Their values were systematically explored within the range of 1–50. For each parameter combination, the model was evaluated using $F1_{\text{mi}}$ and as the model size. These parameters have a significant impact on both the model size and recognition accuracy [66], [67]. Selecting the optimal configuration ensured that the model remained compact while maintaining good recognition performance. The same hyperparameter tuning procedure was applied to all the ML models in Table V, using the same training data and evaluated on the same testing data.

The implementation of embedding the RF model into a microcontroller is detailed, ensuring that the hardware's memory and computational limitations are effectively managed.

1) *Model Conversion and Export:*
   a) Import the model conversion library using: `from micromlgen import port.`
   b) Convert the trained model (clf) into C code: `model_code = port(clf).`
   c) Save the converted model as a header file: `with open("random_forest.h", "w") as file: file.write(model_code)`
2) *Model Integration:*
   a) Embed the header file `random_forest.h` into the microcontroller firmware: `#include "random_forest.h".`
   b) Initialize the RF model in the microcontroller using: `Eloquent::ml::port::RF rf;.`
3) *Deployment and Execution:*
   a) For each input feature vector (feat_j): $\hat{L}_j$ =`rf.predict(feat`$_j$`);`

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Algorithm 1 Hyperparameter Optimization for ML

Input: Full dataset $\mathcal{D}$, model size limit = 1 MB
Output: Optimized parameters ($n_estimators^*, max_depth^*$) and corresponding RF model
Split $\mathcal{D}$ into $\mathcal{D}_{train}$ (60 %) and $\mathcal{D}_{test}$ (40 %) ;
Initialize $best_score \leftarrow 0$, $best_param \leftarrow$ None ;
for *max_depth in [1,2,..,50]* **do**
  for *n_estimators in [1,2,..,50]* **do**
    Train RF model clf on $\mathcal{D}_{train}$ with current parameters ;
    Evaluate clf on $\mathcal{D}_{test}$, compute $score_{val} = f1_{mi}$;
    if $model_size(clf) < 1\ MB$ **then**
      Perform fivefold cross-validation on $\mathcal{D}$ to obtain $score_{cv}$;
      if $|score_{val} - score_{cv}| \leq \varepsilon$ ***and*** $score_{val} > best_score$ **then**
        $best_score \leftarrow score_{val}$;
        $best_param \leftarrow (n_estimators, max_depth)$;
        $best_model \leftarrow clf$;
Export $best_model$;

IV. Experiments and Results

A. Evaluation Method

1) Accuracy (acc) represents the ratio of correct predictions to the total number of predictions

$$\text{acc} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{TN} + \text{FN}}. \tag{4}$$

2) Sensitivity (sen) indicates the model's ability to correctly detect activities that are actually occurring

$$\text{sen} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{5}$$

3) Specificity (spe) shows the model's ability to correctly identify activities that are not occurring

$$\text{spe} = \frac{\text{TN}}{\text{TN} + \text{FP}}. \tag{6}$$

4) Precision (pre) represents the proportion of correctly identified positive activities out of all activities predicted as positive

$$\text{pre} = \frac{\text{TP}}{\text{TP} + \text{FP}}. \tag{7}$$

1) $F1_{\text{mi}}$ is particularly suitable for imbalanced data and $F1$-score macro ($F1_{\text{ma}}$) evaluates the average performance across all activities

$$F1 = \frac{2 \times \text{pre} \times \text{sen}}{\text{pre} + \text{sen}} \tag{8}$$

$$F1_{\text{mi}} = \frac{1}{n} \sum_{i=1}^{n} \frac{2 \times \text{pre}_i \times \text{sen}_i}{\text{pre}_i + \text{sen}_i} \tag{9}$$

$$F1_{\text{ma}} = \frac{1}{n} \sum_{i=1}^{n} F1_i. \tag{10}$$

In (4)–(10), TP—true positive—is the number of times the model correctly predicted that an activity was actually occurring; FP—false positive—is the number of times the model incorrectly predicted that an activity was occurring when it was not; TN—true negative—-is the number of times the model correctly predicted that an activity was not occurring; FN—false negative—is the number of times the model incorrectly predicted that an activity was not occurring when it actually was; and n is the number of activities.

1) K-fold cross-validation: Dataset $\mathcal{D}$ is partitioned into k folds $\mathcal{D}_1, \mathcal{D}_2, \ldots, \mathcal{D}_k$. $k = 5$ is applied in this study.
2) For each fold $\mathcal{D}_k$: Train the model with optimized parameter in Table V on the training set $\mathcal{D}_{\text{train}}^{(k)} = \{\mathcal{D} \setminus \mathcal{D}_k\}$.
3) Compute $F1_{\text{mi}}^k$ for fold $\mathcal{D}_k$.
4) Overall performance: $\text{Score}_{\text{CV}} = (1/k) \sum_{i=1}^{k} F1_{\text{mi}}^k$.

B. Experimental Evaluation

RFAR was experimentally deployed on volunteers using a wearable device integrated with the proposed model during a drill at the University of Fire Prevention and Fighting in Vietnam.

1) Results on Private Dataset: Table V presents the optimal parameter configurations aimed at achieving high $F1_{\text{mi}}$ value while ensuring memory size under 1-MB constraints for recognition models. These models were implemented using the scikit-learn library to maintain consistency throughout the training and evaluation processes. As shown in Fig. 3, the parameters of each algorithm must be properly constrained to ensure that ML remains under 1 MB in size. XGB [Fig. 3(d)] is more suitable for a wide configuration range with max_depth: 1–50 and n_estimators: 1–40, followed by GBDT [Fig. 3(e)], which uses n_estimators: 1–50 and max_depth below 10. In contrast, RF [Fig. 3(a)] and LBM [Fig. 3(b)] adopt more limited parameter ranges to meet the memory constraints. However, RF, XGB, and LBM obtained $F1_{\text{mi}}$ over 95% with optimal parameters. DT [Fig. 3(c)] and SVM [Fig. 3(f)] are more compact, with all configurations under 1 MB, but their $F1_{\text{mi}}$ values are lower than those of RF (96.1%), XGB (95.5%), and LBM (95.2%).

Under memory constraints, RF demonstrated a good balance among high accuracy, model size, and processing speed. With a configuration of n_estimators = 17 and max_depth = 14, the model occupied only 0.921709 MB while achieving acc = 99.3% and over 96% across all the metrics. In addition, RF provided fast processing with a training time of 0.093589 s on 4171 windows and an inference time of 0.010951 s on 2774 windows, corresponding to approximately 3.95 μs per prediction. In comparison, models such as XGB (n_estimators = 37, max_depth = 14, 0.939840 MB) and LBM (n_estimators = 46, max_depth = 22, 0.966480 MB) achieved similar accuracy (above 99%) but required more trees and longer training times (XGB: 0.803459 s, LBM: 0.226004 s). Although DT achieved the shortest inference time

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TABLE V
CONFIGURATION PARAMETERS OF RECOGNITION MODELS ON PRIVATE DATASET

Model	Hyperparameters	acc (%)	sen (%)	spe (%)	pre (%)	$F1_{mi}$ (%)	$F1_{ma}$ (%)	Size (MB)	Train (s)	Test (s)
RF	[1]n_estimators = 17, [2]max_depth = 14	99.3	97.0	99.6	97.1	96.1	97.0	0.921709	0.093589	0.010951
XGB	n_estimators = 37, max_depth = 14	99.2	96.4	99.5	96.6	95.5	96.5	0.939840	0.803459	0.047790
DT	max_depth = 16	98.5	93.3	99.1	93.6	91.7	93.5	0.056939	0.206093	0.000996
GBDT	n_estimators = 35, max_depth = 4	98.8	94.5	99.3	94.9	93.4	94.7	0.967428	21.020377	0.014934
SVM	[3]$C_ = 29$	98.9	94.8	99.4	94.6	93.7	94.7	0.247342	21.748170	0.091596
LBM	n_estimators = 46, max_depth = 22	99.1	96.1	99.5	96.4	95.2	96.2	0.966480	0.226004	0.007965

[1] Number of trees in boosting models.
[2] Maximum depth of each DT in the model.
[3] Regularization parameter in SVM.

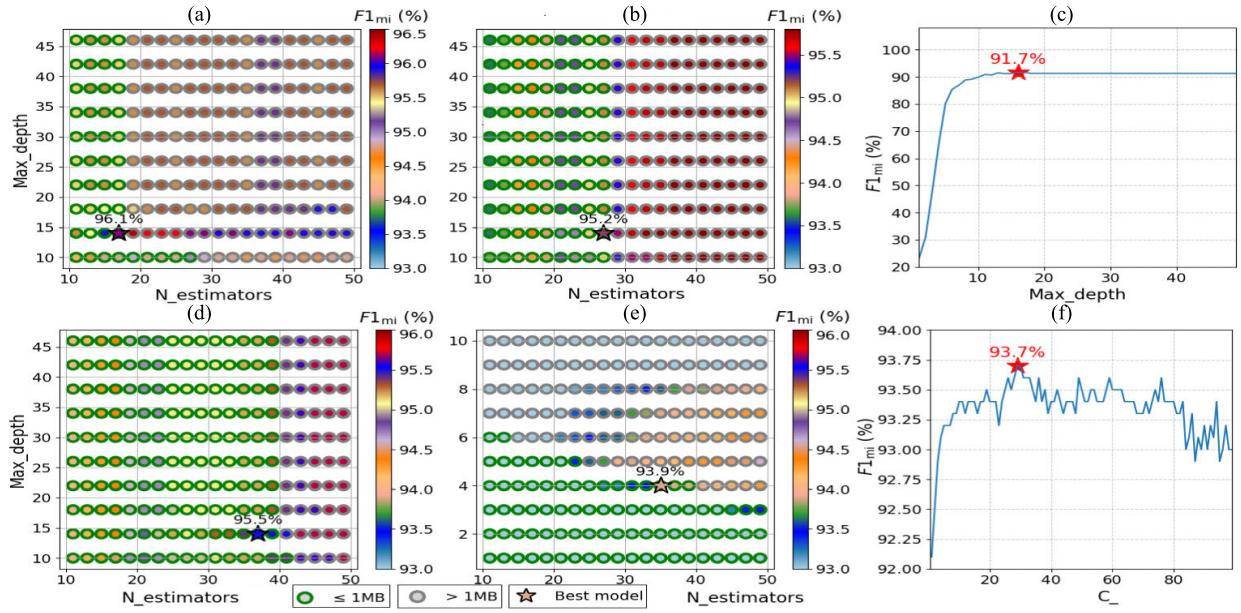

Fig. 3. $F1_{mi}$ and model size of models (a) RF, (b) LBM, (c) DT, (d) XGB, (e) GBDT, and (f) SVM when tuning parameters.

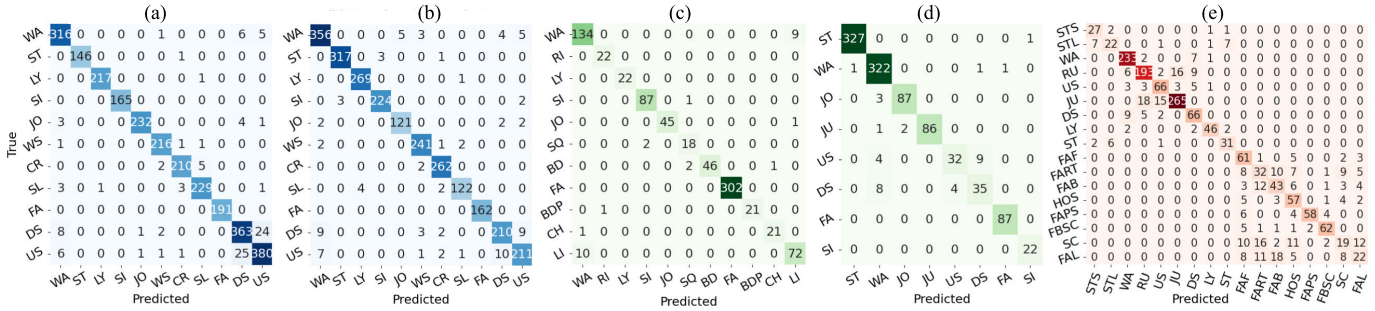

Fig. 4. Confusion matrix summarizes the results on (a) private dataset, (b) real-time experiments on volunteers, (c) SFDLA, (d) MobiFall, and (e) UniMiB.

(0.000996 s) with a very small model size (0.056939 MB), its metrics remained above 91%.

Fig. 4(a) illustrates the confusion matrix on the private dataset. The RF model correctly recognized most activities, such as WA (316/321), ST (146/148), LY (217/219), SI (165/166), and JO (232/235), reflecting clear separability in their signal patterns. However, notable confusions were observed between similar locomotion activities. Specifically, 24 DS were misclassified as US, and five WA as US. These errors likely stem from overlapping motion features during elevation changes or transitions, where acceleration patterns are similar. For instance, DS↔US and WS↔WA involve actions with comparable gait profiles but differing intensity or direction. CR and SL were generally recognized correctly. However, they showed slightly more confusion with similar low-speed or transitional actions such as WS and LY. This may be caused by signal variation or short data windows. In contrast, static postures like ST and LY were recognized with high accuracy due to low variation within the class. Most misclassifications occurred between activity pairs that share similar movement patterns.

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TABLE VI
ACTIVITY EVALUATION METRICS FROM (a) PRIVATE DATASET AND (b) REAL-TIME EXPERIMENT

Data	Activity	acc (%)	sen (%)	spe (%)	pre (%)	$F1$ (%)
(a)	WA	98.8	96.3	99.1	93.8	95.0
	ST	100	99.3	100	100	99.7
	LY	99.9	99.5	100	99.5	99.5
	SI	100	100	100	100	100
	JO	99.6	96.7	99.9	99.1	97.9
	WS	99.7	98.6	99.6	97.3	98.0
	CR	99.5	96.8	99.8	97.2	97.0
	SL	99.5	96.6	99.7	97.0	96.8
	FA	100	100	100	100	100
	DS	97.5	91.2	98.5	91.2	91.2
	US	97.7	91.8	98.7	92.5	92.1
	Overall	99.3	95.9	99.6	96.4	96.1
(b)	WA	98.6	95.4	99.1	94.7	95.1
	ST	99.7	98.8	99.9	99.1	98.9
	LY	99.8	99.6	99.8	98.5	99.1
	SI	99.7	97.8	99.9	98.7	98.2
	JO	99.6	95.3	99.8	96.0	95.7
	WS	99.5	98.0	99.6	96.4	97.2
	CR	99.6	99.2	99.7	97.0	98.1
	SL	99.6	95.3	99.8	96.8	96.1
	FA	100	100	100	100	100
	DS	98.5	90.1	99.3	92.9	91.5
	US	98.5	90.9	99.2	92.1	91.5
	Overall	99.4	96.4	99.6	96.6	96.2

Table VI summarizes the performance metrics of the proposed method across 11 activities, evaluated on the private dataset and real-time experiments. On the private dataset [Table VI(a)], activities were recognized with high and stable performance across evaluation metrics, with minor variations among actions. Actions such as ST, SI, and FA reached 100% in acc, sen, spe, pre, and $F1$. Other activities, including LY, SL, JO, CR, and WA, also achieved high values, with most metrics exceeding 95%. In contrast, US and DS showed relatively lower results. Specifically, US achieved sen $= 91.8\%$ and $F1 = 92.1\%$, while DS reached sen $= 91.3\%$ and $F1 = 91.2\%$. However, their acc and spe remained above 97%.

A detailed comparison of ML models after hyperparameter tuning on the private dataset is presented in Table V. RF achieved high values across all the evaluation metrics, with acc reaching 99.3%, and sen, spe, pre, $F1_{\text{mi}}$, and $F1_{\text{ma}}$ all around 97%. For acc, these models achieved high values above 99%, suggesting that extracted features are highly discriminative across activities. In terms of sen, RF recorded the highest values (97%–97.2%), followed by XGB and LBM (around 96.6%). DT and SVM showed slightly lower sen (95.4%–95.9%), which may affect the detection of less distinctive activities. Regarding spe, these models achieved similarly high levels between 99.4% and 99.6%, suggesting reliable rejection of negative cases. For pre, RF achieved 97.1%, while XGB and LBM maintained competitive results at 96.6%. DT and SVM yielded pre (95.4%–95.5%), indicating a marginally higher false positive rate.

Overall, all the models exhibited consistent performance across different evaluation settings, as reflected in their stable results across all the evaluation metrics. RF maintained relatively balanced scores, while XGB and LBM also achieved stable and comparable outcomes. Although DT, GBDT, and SVM produced lower values, their results remained steady without notable variations. The consistency across both the evaluation schemes indicates that the models generalized well, with no indication of overfitting.

TABLE VII
COMPARISON OF THE PERFORMANCE OF RF ON PUBLIC DATASETS

Dataset	Size (MB)	acc (%)	sen (%)	spe (%)	pre (%)	$F1_{mi}$ (%)
SFDLA	0.750984	99.4	95.7	99.6	95.9	96.6
MobiFall	0.429473	99.2	92.2	99.5	94.2	96.6
UniMiB	0.975449	97.4	72.6	98.6	72.2	78.2

2) Results of SFDLA Dataset: For the SFDLA dataset, data collected from female participants were excluded. Moreover, sensor data from positions such as the head, right wrist, right thigh, and chest were removed, as these placements were unsuitable for accurately capturing operational movements. After filtering, signals from the waist-mounted sensor were segmented into 3-s windows.

Fig. 4(c) presents the confusion matrix on the SFDLA dataset. The RF model achieved high accuracy across most activity classes. Actions such as RI, LY, and FA were classified without errors in the current test set. Other activities, including SI (87/88), JO (45/46), squatting (SQ) (18/20), bending (BD) (46/49), BD-pickup (BDP), and 21/22 coughing-sneezing (CH), also showed high recognition rates.

WA and LI, which exhibit similar movement patterns, were occasionally misclassified. Specifically, nine WA were predicted as LI, and ten LI were predicted as WA. This bidirectional confusion suggests that the two actions share similar motion characteristics, particularly in gait-related patterns. Their differences may be limited in short observation windows or under certain execution styles. A small number of misclassifications occurred in other classes. For instance, one BD instance was predicted as CH. This may be due to a brief forward-leaning posture that appears in both the actions during certain time windows.

The overall performance of the proposed RF model on the SFDLA dataset is reported in Table VII, with acc $=$ 99.4%, sen $=$ 95.7%, spe $=$ 99.6%, pre $=$ 95.9%, and $F1_{\text{mi}} = 96.6\%$. These results indicate that most activities were correctly detected, false positives were limited, and the balance between sensitivity and precision was maintained across classes.

3) Results of MobiFall Dataset: The MobiFall dataset includes four types of falls: forward-lying (FOL), front-knees-lying (FKL), back-sitting-chair (BSC), and sideward-lying (SDL); as well as nine daily activities: ST, WA, JO, jumping (JU), US, DS, SI, car-step in (CSI), and car-step out (CSO). This study focuses on male subjects performing rescue operations inside a burning building. Therefore, certain activities such as BSC, CSI, and CSO are excluded from the analysis. Activities recorded from female volunteers are also not considered.

Recognition performance on the MobiFall dataset is illustrated in Fig. 4(d), with detailed evaluation metrics summarized in Table VII. RF recognized most actions with high accuracy, such as ST (327/328), WA (322/325), JO (87/90), JU (86/89), SI (22/22), and FA (87/87). This result is consistent

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with the observation from the SFDLA dataset, where ST and WA actions also showed high correctness.

However, the recognition rate decreased for activities with similar posture, such as US (32/45) and DS (35/47). These actions involve stair movement, which shares similar lower limb motion with WA and JU. The confusion may result from overlapping acceleration patterns during climbing and descending. Misclassifications between US and DS also occurred, indicating that elevation-related activities remain difficult to distinguish using only inertial data.

Most activities were correctly detected, and false detections were limited, despite challenges in distinguishing stair-related actions. RF achieved acc = 99.2%, sen = 92.2%, spe = 99.5%, pre = 94.2%, and $F1_{\mathrm{mi}}$ = 96.6%. In summary, the proposed model with optimized parameters reached a good balance on the MobiFall dataset.

4) Results of UniMiB Dataset: For the UniMi dataset, 17 activities (AF17) were considered, including StandingUpFS (STS), StandingUpFL (STL), WA, Running (RU), US, JU, DS, LyingDownFS (LY), SittingDown (SI), FallingForw (FAF), FallingRight (FAR), FallingBack (FAB), HittingObstacle (HOS), FallingWithPS (FAPS), FallingBackSC (FASC), Syncope (SC), and FallingLeft (FAL). A major challenge of this dataset is that the smartphone was placed in the front pocket, and its orientation varied significantly between data collection sessions, including cases where the device was inverted or rotated, which made HAR more difficult. Therefore, the actions were preprocessed and mapped to a common orientation when similar activities began.

These activities were categorized into two main groups: 1) normal activities of daily living (ADL), which included STS, STL, WA, RU, US, JU, DS, LY, and ST, and 2) fall-related activities. The recognition results on the UniMiB dataset are presented in Fig. 4(e), and the corresponding evaluation metrics are provided in Table VII. Some activities reached high recognition accuracy, such as WA with 233/242 (96.3%), LY with 46/52 (88.5%), and RU with 193/229 (84.2%). In addition, transitional activities such as STS (27/31, 87.1%), US (66/80, 82.5%), and DS (66/82, 80.5%) were also well-recognized. STL was the most difficult to classify, only reaching 22/38 (57.9%). Most STL samples were misclassified as STS or ST, which have similar starting or ending postures.

In contrast, the fall-related activities' group, consisting of eight classes with a total of 879 windows, showed low recognition performance, with only 431 windows correctly recognized. Fall activities were mostly confused with each other. Some actions, such as FAF (61/72, 84.7%), FAPS (58/72, 80.6%), and FBSC (62/78, 79.5%), had relatively good recognition accuracy. However, other activities, such as FART (32/79, 40.5%), FAL (22/76, 28.9%), and especially SC (19/85, 22.4%), showed low recognition accuracy. These falls produced similar acceleration changes at the moment of impact, which reduced the model's ability to separate them.

With sen and pre only at 71%–72% and $F1_{\mathrm{mi}}$ averaging 78.2%, the model still had difficulty in recognizing different types of falls (Table VII). Compared with the SFDLA and MobiFall datasets, where the model achieved high recognition rates across most activities, UniMiB posed more challenges due to the high number of fall classes and the similarity between them. However, the proposed model was able to accurately distinguish between ADLs and fall-related activities, without misclassifying falls as ADLs. This matches the proposed model's ability to correctly recognize dangerous states such as falls.

Fig. 5. Experimental process on firefighters during rescue and emergency drills.

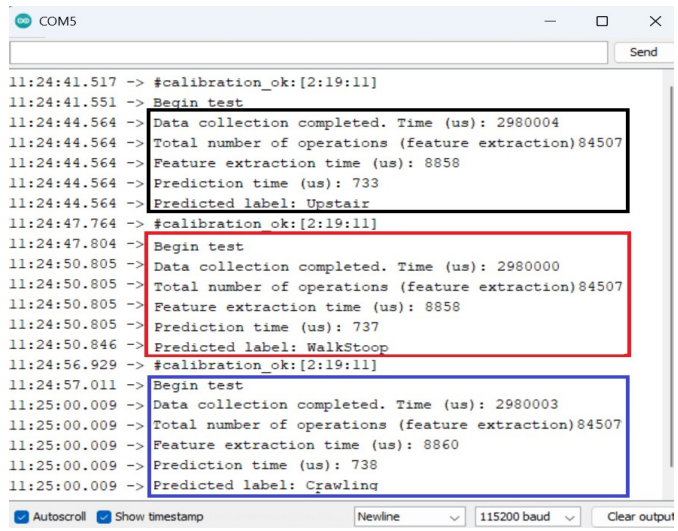

Fig. 6. Deployment results of prediction on the microcontroller.

C. On-Device Deployment and Practical Evaluation

1) Volunteer Evaluation Setup: During the drill, the volunteers performed 11 activities under various conditions and repeated each activity multiple times (Fig. 5). The recognition results are presented in Fig. 4(b), along with the evaluation metrics summarized in Table VI(b). In addition, a demo video is provided in the Appendix to illustrate the proposed model's real-time recognition capability after deployment on a microcontroller. Some actions may exhibit slight display delays due to signal transmission and reception errors.

2) Inference Time, Latency, and Computational Complexity: In this study, the step size (sliding window step) equals the interval between two consecutive observations, which determines the time between recognition instances. Under normal conditions, the step size is set equal to the window size (3 s), so recognition is performed every 3 s to minimize energy consumption. When the AIx value exceeds a predefined threshold, the step size is reduced to 0.5 s to increase the recognition frequency, while the window size remains constant at 3 s.

Fig. 6 shows the actual and computation times on the microcontroller for each processing stage, including data

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collection, feature extraction, and recognition. With the wearable device integrating an ESP32 microcontroller and an ADXL345 accelerometer, the end-to-end processing flow was evaluated to ensure real-time capability. During operation, the accelerometer samples three-axis data at 50 Hz, and the system collects 450 samples within a 3-s window. Once data collection is complete, feature extraction is performed with approximately 84 507 lines, requiring about 8858 μs. The proposed model, with 140 838 instruction lines, completes inference in approximately 733–738 μs. Following recognition, the recognition result is transmitted via SPI at 10 MHz, which takes only about 1.6 μs and is negligible.

The proposed model used in this pipeline is implemented directly on performance-constrained microcontrollers. It is configured as an RF with 17 DTs, each having a maximum depth of 14, and has a compiled size of approximately 0.92 MB. This model has a recognition latency of approximately 9.561–9.596 ms on the ESP32 microcontroller, which is better than that reported in some related studies [25], [50], [63], [65], [68] on TinyML applications for microcontrollers.

3) Real-Time Recognition Results: The real-time recognition performance for actions during the drill is presented in Table VI and Fig. 4(b). The model achieved acc = 99.4%, sen = 96.4%, spe = 99.6%, pre = 96.6%, and $F1 = 96.2\%$. The FA action was correctly recognized in all 162 windows, with 100% for all the evaluation metrics. Actions such as ST (317/320), LY (269/270), SI (224/229), WS (241/246), JO (121/127), CR (262/264), and SL (122/128) were also recognized with high accuracy. All the corresponding metrics exceeded 95%. In contrast, WA had 17 misclassifications, mostly as JO, WS, DS, or US. DS and US also had 23 and 21 incorrect predictions, respectively. Similarly, 23/233 DS and 21/232 US were also misclassified. In particular, DS and US recorded sen, pre, and $F1$ just above 90%. The performance degradation for WA, DS, and US is likely due to the similar postures of these activities, which pose challenges for accurate real-time classification with short window sizes. Nevertheless, the number of misclassifications was relatively small and considered acceptable. The real-time results were comparable to those on the private dataset and MobiFall, with all the main metrics above 96%. Posture transitions and static actions were consistently recognized, as also observed in SFDLA and MobiFall. In contrast, UniMiB included more fall types with similar motion patterns, which made recognition more difficult than in the real-time setting.

In general, the embedded model running on the microcontroller achieved results similar to those of the model tested on the private dataset. Both the models showed less than 1% difference in acc, sen, spe, pre, and $F1_{\text{mi}}$. Specifically, the model tested on the private dataset reached acc = 99.3% and $F1_{\text{mi}} = 96.1\%$, while the real-time experiment achieved acc = 99.4% and $F1_{\text{mi}} = 96.2\%$. Marginal differences were also observed in sen (95.9% and 96.4%) and pre (96.4% and 96.6%). These small differences indicate that recognition performance remained stable when the model was applied in practical conditions. This outcome is attributed to the use of an anomaly detection algorithm, which enables the proposed model to more effectively handle posture transitions, thereby reducing noise in practical activity sequences. In addition, since the volunteers had body shapes similar to those of firefighters, variations in sensor orientation were also minimized.

TABLE VIII
COMPARISON OF RECOGNITION PERFORMANCE ON THE UNIMIB-SHAR DATASET (a) AF17 AND (b) AF2

Dataset	Study	Acc (%)	$F1_{mi}$ (%)
(a)	Proposed	97.4	78.2
	Xu et al. [37]	97.6	79.7
	Teng et al. [52]	97.2	78.6
(b)	Proposed	100	100
	Al-qaness el al. [69]	99.95	–
	Kanjilal el al. [70]	99.82	98.86
	Iloga at al. [71]	98.85	–

V. DISCUSSION

A. Performance of Proposed Model

Table VIII shows comparison of the proposed method with previous studies on the UniMiB-SHAR dataset, based on two activity groups: (a) AF17 and (b) AF2. For AF17, the proposed model achieved acc = 97.4% and $F1_{\text{mi}} = 78.2\%$, which are lower than those of CNN-DCT [37] (acc = 97.6%, $F1_{\text{mi}} = 79.7\%$) but higher than those of CNN-TSFDU-LW [52] (acc = 97.2%, $F1_{\text{mi}} = 78.6\%$). CNN-DCT is capable of learning frequency-domain features, whereas CNN-TSFDU-LW uses the temporal–spatial feature decoupling unit (TSFDU) to separate essential features to reduce computational complexity. Both the approaches demonstrate strengths in capturing signal characteristics of each activity. However, the accuracy differences between the proposed model and these two models are relatively small. This outcome was achieved due to careful preprocessing, which normalized the input signals from the smartphone into a unified reference frame. For example, signals corresponding to activities such as STS, RU, WA, and FA were aligned to a common direction representing the upright standing posture.

The proposed model demonstrated good and stable recognition performance on the private dataset, with both acc and $F1_{\text{mi}}$ exceeding 96%. For public datasets, MobiFall and SFDLA achieved comparable results ($F1_{\text{mi}} = 96.6\%$). However, UniMiB performed significantly lower, with $F1_{\text{mi}} = 78.2\%$. This is mainly due to data collection with varying phone orientations in UniMiB, which increases the challenge for recognition. In addition, the proposed algorithm is well-suited for recognizing firefighter-specific actions, which require frequent posture changes and may involve falls. However, distinguishing specific types of falls remains limited. Furthermore, the device stability considerably affects recognition performance.

If fall is distinguished from routine daily activities as nonfall instances, the method achieves 100% in sen, spe, acc, and GM (GM = $(\text{spe} \times \text{sen})^{1/2}$). Table IX presents a comparison of the proposed fall detection method with previous studies. Compared with previous studies, the method shows clear improvements. For example, Pham et al. [11] reported sen of 95.89%, spe of 100%, acc of 97.96%, and GM of 95.24%. In addition, results from Dang et al. [33] showed

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TABLE IX
COMPARISON OF THE PROPOSED FALL DETECTION MODEL WITH PREVIOUS OTHER MODEL FOR FIREFIGHTER

Study	*sen* (%)	*spe* (%)	*acc* (%)	*GM* (%)
Proposed	100	100	100	100
Dang et al. [33]	100	98.88	99.49	99.40
Chai et al. [29]	92.25	94.59	94.10	93.40
Thanh et al. [11]	95.89	100	97.96	95.24

TABLE X
COMPARISON OF THE PROPOSED FALL DETECTION MODEL WITH PREVIOUS STUDIES ON (a) SFDLA AND (b) MOBIFALL DATASETS

Dataset	Study	*sen* (%)	*spe* (%)	*acc* (%)	*GM* (%)
(a)	Proposed	100	100	100	100
	Dang et al. [33]	95.89	100	97.96	97.92
	Thanh et al. [11]	93.33	91.67	92.50	92.50
	Enol et al. [72]	100	99.97	99.97	99.98
(b)	Proposed	100	100	100	100
	Yi et al. [36]	100	100	100	100
	Zhang et al. [73]	100	–	99.65	–
	Dang et al. [33]	100	98.88	99.49	99.44
	Hussain et al. [74]	96.72	–	98.60	–

sen of 95.89%, spe of 100%, GM of 99.4%, and acc of 99.49%, while the proposed model demonstrates superiority by achieving a better balance between correctly detecting falls and minimizing false positives. The methods proposed by Thanh and Dang share a common limitation in that they rely on fixed signal thresholds, which require manual tuning and are less adaptable to individuals with varying body shapes. Meanwhile, Chai et al. [29] obtained lower performance with sen of 92.25%, spe of 94.59%, acc of 94.1%, and GM of 93.4%. The main cause is primarily attributed to the sensor placement on the chest, which increases the likelihood of misclassifying FA and WS. This limitation is effectively addressed in our study through an anomaly detection mechanism that reduces false fall signals during postural transitions or activity changes by the participants.

Table X presents a comparative evaluation of the proposed fall detection model with the existing approaches on the SFDLA and MobiFall datasets. The proposed method achieves consistently high recognition results on the MobiFall, SFDLA, and UniMiB datasets. On the MobiFall dataset, the proposed model achieves sen, spe, acc, and GM all at 100%, surpassing previous studies such as Zhang et al. [73], which also achieved 100% in sen. Compared with studies such as Pham et al. [11], the proposed model shows significant improvements in sen (100%, 90.16%, and 100%) and acc (100%, 96.16%, and 99.49%). Although both the proposed model and Yi's model achieve an acc of 100%, Yi's model relies on 36 features and a complex LSTM architecture, which consumes significantly more resources. Therefore, the proposed approach offers a simpler and more efficient solution for practical applications.

On the SFDLA dataset, the proposed model also maintains maximum performance with all the metrics reaching 100 %, outperforming prior studies such as Dang et al. [33] with an acc of 97.96% and GM of 97.92%, and Pham et al. [11] with sen of only 93.33%. The difference in the results between these two public datasets is attributed to the placement of the data collection device. In the MobiFall dataset, the device used for data collection was a smartphone placed in a pocket near the waist, while in the SFDLA dataset, the device is securely fixed at the waist. This can lead to the SFDLA data being more reliable and less prone to noise compared with the MobiFall dataset.

For AF2 [Table VIII(b)], the proposed method achieved 100% recognition accuracy for hazardous fall, consistent with the results on other datasets as SFDLA and MobiFall. In general, models such as CNN-PCNN [69], 1D-CNN [70], and HMMs [71] also achieved very high accuracy for fall actions (over 98%). However, the proposed model maintained reliable accuracy while avoiding misclassification between fall actions and ADLs. These models have demonstrated strong capabilities in fall detection and activity recognition, particularly in their ability to automatically extract discriminative features from raw sensor data. However, the proposed model incorporates a processing flow that improves fall detection accuracy by mitigating misclassification caused by transitional or ambiguous movements that resemble falls during daily activity recognition.

B. Sliding Window and Sampling Frequency

The choice of window size and F_s has a major influence on feature extraction and the optimization of ML performance [75]. This impact on $F1_{\text{mi}}$ score is illustrated in Fig. 7, where six models were evaluated on the same private dataset. The experimental results show that the 6-s (overlap 50%) window, when applied to models, achieves stable and notable effectiveness across the entire F_s range from 20 to 100 Hz. The models using the proposed feature set achieved $F1_{\text{mi}}$ scores above 95% when operating at sampling frequencies between 50 and 100 Hz with window sizes ranging from 3 to 6 s. While high performance was observed at 50 Hz, increasing the sampling rate to 100 Hz resulted in only marginal improvements, which were not significant considering the additional resource and energy consumption. For instance, RF, LBM, GBDT, DT, SVM, and XGB exhibited increases of only 0.2%–0.3% when changing from 50 to 100 Hz, yet the maximum recognition performance could not be fully reached. A 6-s window yielded high performance, with $F1_{\text{mi}}$ scores above 99% (e.g., GBDT and LBM); however, reducing the window to 3 s led to a decrease of approximately 2%–3%, and further shortening it to 1 s caused a more substantial drop of about 15%.

Although the 6-s window yielded $F1_{\text{mi}}$ above 99%, it posed a challenge for recognition of firefighter activities, where multiple actions often occur simultaneously within an observation. In contrast, shorter windows of 1 and 2 s showed a significant performance reduction. The 3-s window showed a practical balance. Although its $F1_{\text{mi}}$ was approximately 2%–3% lower than that of the 6-s window, it significantly mitigated the issue of overlapping actions in single observations and also reduced resource requirements for practical deployment. In addition, an F_s of 50 Hz was found to be appropriate, as it reduced energy consumption compared with 100 Hz while still maintaining reliable recognition accuracy.

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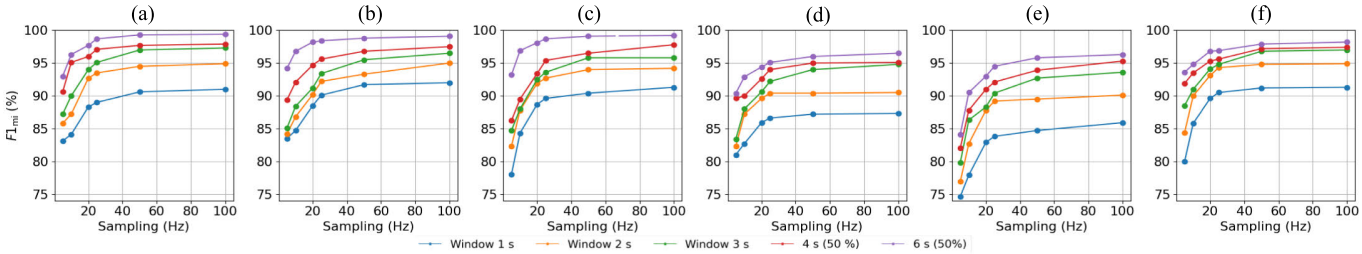

Fig. 7. $F1_{\text{mi}}$ of (a) RF, (b) XGB, (c) GBDT, (d) SVM, (e) DT, and (f) LBM, based on changes in F_s and window size.

TABLE XI
COMPARISON OF FEATURE EFFECTIVENESS ON THE PRIVATE DATASET

Applied features	Feature types	$F1_{mi}$ (%)
Proposed	13	**96.1**
Remove energy	12	93.4 (↓2.7)
Remove mean absolute deviation	12	93.5 (↓2.6)
Remove signal magnitude area.	12	93.3 (↓2.8)
Remove average amplitude change	12	93.0 (↓3.1)
Remove simple square integral	12	93.8 (↓2.3)
Remove mean	12	92.6 (↓3.5)
Remove standard deviation	12	93.0 (↓3.1)
Remove maximum	12	93.0 (↓3.1)
Remove interquartile range	12	93.6 (↓2.5)
Remove hjorth complexity	12	93.0 (↓3.1)
Remove median absolute deviation	12	93.8 (↓2.3)
Remove range	12	93.2 (↓2.9)
Chai [30] proposed 20 time and 6 frequency domain feature types	26	95.6 (↓0.6)
Wang [35] proposed 11 time and 2 frequency domain feature types	16	92.9 (↓3.2)
Vi [25] proposed average absolute deviation, std, iqr, range, and energy	5	85.7 (↓10.4)
Hong [22] proposed mean, std, median, range, energy	5	83.1 (↓13)

C. Optimizing Features

Selecting features based on their importance has effectively reduced the number of features [60], [61], computational complexity [22], [25], and model size, while maintaining high performance [24], [26]. In this study, the combination of 13 feature types provides comprehensive insights into signal oscillation, symmetry, and stability, significantly enhancing the accuracy of action classification. Similarly, previous studies demonstrated that feature selection significantly impacted the accuracy of recognition models [30], [32], [35].

Table XI shows comparison of $F1_{\text{mi}}$ performance when modifying the proposed feature set. Adjusting the number of features compared with the proposed feature set has resulted in poorer performance. When the full set of proposed features (13 features) was applied, the model achieved $F1_{\text{mi}} = 96.1\%$, which was the highest among all the cases. When individual feature types were removed, $F1_{\text{mi}}$ decreased in all the cases, indicating that each feature contributed to recognition performance. In particular, removing features such as energy, mean absolute deviation, or signal magnitude area led to more notable drops in performance, with $F1_{\text{mi}}$ reduced to 93.4%, 93.5%, and 93.3%, respectively. For feature sets with larger numbers of features, such as Chai's set of 26 feature types [30], an $F1_{\text{mi}}$ of 95.6% was achieved, while Wang's 16 feature types [35] reached 92.9%. In contrast, feature sets with fewer features, such as those proposed by Vi [25] and Hong et al. [22], resulted in lower performance, with $F1_{\text{mi}}$ values of 85.7% and 83.1%, respectively. Therefore, the proposed feature set offers a good balance between feature quantity and recognition performance.

In general, the proposed feature set contains comprehensive information, achieving a balance between classification performance and computational complexity, making it suitable for specific requirements of firefighter action recognition. While other feature sets may reduce the number of features or increase the processing speed, they fail to compensate for the loss of critical information when certain features are removed. In the context of prolonged rescue operations, increasing the number of features results in higher computational time, negatively affecting recognition performance when processing real-time data, while the improvement in accuracy is negligible, or accuracy may even decrease.

D. Limitation and Feature Work

Although the proposed system has many advantages, several limitations need to be addressed to enhance its practical applicability. First, the system has a large size due to its manually assembled design, which causes inconvenience during field deployment. Optimizing its size and design for easy integration with support equipment such as clothing and belts is essential. A further limitation is that the system was developed and evaluated primarily using data from male volunteers with body types closely resembling those of professional firefighters. While this sampling strategy aligns with the system's intended application, it may limit the model's adaptability to firefighters with substantially different physical characteristics, such as female personnel or those with atypical body builds.

In addition, the current system is only a small part of the overall positioning system and has not yet incorporated critical features such as location tracking and emergency alerts. Adding these features will improve usability and facilitate timely information transmission to commanders. However, the integration process requires close coordination among software, hardware, and data transmission protocols, necessitating additional time and resources for development. Optimizing the system's size, weight, and hardware upgrades will be a key objective to provide a more comprehensive and effective solution for firefighters in emergency situations.

In the future, the research aims to develop a comprehensive emergency support system that includes monitoring work status, issuing emergency alerts in hazardous situations, and creating a support network for multiple firefighters simultaneously.

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VI. Conclusion

This study constructed a dedicated dataset simulating firefighter rescue activities to supplement existing public datasets and developed a signal processing pipeline that combines a sliding window technique and AIx to improve the recognition of transition actions. As a result, the position and starting time of the sliding window are adjusted appropriately to fully capture an action. Based on this foundation, the RFAR system, which uses a lightweight wearable device and an optimized RF model, was implemented on a low-power microcontroller. The system achieved an $F1_{\text{mi}}$ of over 96% in private dataset and over 78% on three public datasets. The system operated effectively on real-time accelerometer data from volunteers wearing the device during rescue drills, with all metrics exceeding 96%. In terms of practical applications, the proposed system is highly applicable and serves as a vital component of an indoor emergency positioning support system for firefighters.

Appendix

Demo video showcasing the research: https://youtu.be/S4hAzgEo-NA

Source code: https://github.com/HieuSSALAB/FireFighter

Declarations

Data availability: Dataset in this research was collected by the authors and volunteers, fully complying with privacy and ethical regulations. Scientists wishing to use the data must cite this article and acknowledge the source. Link: https://dx.doi.org/10.21227/9w3p-v638

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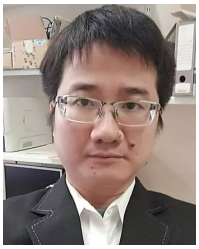

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